

Muhammad Kashif, Ph.D.

Postdoctoral Research Associate (Center for Quantum and Topological Systems) and Research Team Lead (eBRAIN Lab), New York University (NYU) Abu Dhabi, United Arab Emirates

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Google Scholar (Citations: 1100+ , *h-index*: 21, *i-10 index*: 31) — LinkedIn — ORCID

Career Highlights

- 3 years of postdoctoral research experience in quantum computing, quantum algorithms, and quantum machine learning at New York University Abu Dhabi.
- Published 15+ journal articles and 30+ conference papers in leading venues including DAC, DATE, Scientific Reports, EPJ Quantum Technology, QCE, QAI, and IJCNN.
- Developed novel methodologies for quantum algorithm benchmarking, trainability analysis, resource-aware quantum computing, and hybrid quantum-classical architectures.
- Led and contributed to internationally funded research projects spanning quantum machine learning, quantum optimization, and quantum system design.
- Recipient of multiple research awards, including the Best Researcher Award at eBRAIN Lab (NYU Abu Dhabi) and First Prize at the QInnovation World Challenge 2025.
- Supervised and mentored undergraduate and graduate researchers while leading collaborative research activities across academia and industry.

I am a Postdoctoral Research Associate at New York University Abu Dhabi working at the intersection of quantum algorithms, quantum computing systems, and quantum machine learning. My research investigates how quantum computational resources can be effectively utilized to achieve practical advantages in real-world applications through the development, analysis, and benchmarking of quantum algorithms. My expertise includes variational quantum algorithms, quantum optimization, resource-aware quantum computing, trainability analysis of parameterized quantum circuits, quantum advantage assessment and generative AI for quantum programming. By combining theoretical analysis with hardware-aware evaluation, I aim to develop scalable and practical quantum computing solutions for scientifically and societally impactful problems.

Professional Experience

Postdoctoral Research Associate, New York University Abu Dhabi (UAE) Aug 2023–Present

- Developed analytical and hardware-calibrated quantum resource cost models
- Conducted system-level analysis of quantum circuits including noise, connectivity, and compilation effects
- Developed resource-aware neural architecture search frameworks optimizing accuracy–hardware resource cost trade-offs
- Leading research on hardware-aware QML and noise-resilient training of quantum neural networks (QNNs)
- Investigating quantum algorithms (QAOA, VQE etc) for different real world applications (Path planning, Unit commitment, Robot Navigation etc)
- Lead research on trainability enhancement for QNNs (barren plateau mitigation) across feedforward, quantum convolutional and Graph Recurrent QNN architectures
- Developing generative AI approaches for quantum code generation using domain-specific Large Language Models
- Benchmarking QML models for applications, such as finance, classification, Healthcare
- Supervising graduate and undergraduate research projects

Research & Teaching Assistant, Istanbul Şehir University (Turkey) Mar 2018–Dec 2019

- *Teaching*: Conducted undergraduate lab sessions and physics problem solving sessions (60+ students)
- *Research*: FPGA-based cryptographic accelerators (ECC) — design, optimization, and evaluation

Research Engineer (Remote), Collaborative Projects, NSTIP (Saudi Arabia) Feb 2015–Aug 2017

1. *Model-based design and verification of safety-critical embedded systems (SysML/MARTE)*: The project objective was to develop a holistic model-based development methodology for embedded systems, where the system is designed at an abstract level and then automatically transformed to the low level. **My Contribution** was the development of structural and behavioral models of various embedded systems in SysML like, Unmanned Aerial Vehicle, Intelligent Traffic Controller, Car Collision Avoidance System

2. *Hardware implementation and optimization of FPGA-based security algorithms*: The objective was to develop ECC-based crypto-processors with less hardware resources and high throughput. **My Contribution** was hardware implementation of crypto processors and their optimization in terms of area and speed
3. *Temporal Verification of Real-Time Systems*: The project objective was to reduce the computational cost of the temporal testing process where temporal testing time is reduced by predicting the execution times rather than actually running the application program. **My Contribution** was to implement different sorting algorithms in C language, which were then feeded to the genetic algorithm in Matlab

Education

Ph.D., Computer Science and Engineering, Hamad Bin Khalifa University (Qatar)	2020–2023
Thesis: <i>Quantum Machine Learning on Noisy Intermediate-Scale Quantum Devices</i>	
M.Sc., Electronics and Computer Engineering, Istanbul Şehir University (Turkey)	2018–2019
Thesis: <i>A hardware-efficient elliptic curve accelerator for FPGA-based cryptographic applications</i>	
B.Sc., Electrical (Electronics) Engineering, COMSATS (Pakistan)	2011–2015
FYP: <i>Home Automation System using Raspberry Pi over Wi-Fi</i>	

Teaching and Student Supervision

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- Teaching and mentoring experience in different universities, including Istanbul Şehir University Turkey and New York University Abu Dhabi, UAE
 - Undergraduate and graduate education, including lab and problem solving sessions, and hands-on workshops
 - *M.Sc. Thesis Co-Supervision*: Noise Mitigation Techniques for Reliable Quantum Neural Networks, Université Libre de Bruxelles (2024–2025)
 - *B.Sc. Thesis co-Supervision*: Comparative Analysis of Noise and Robustness Evaluation in Quantum Neural Networks, New York University Abu Dhabi (2023–2024)
 - Mentor, NYUAD Quantum Hackathon for Social Good (2024,2025,2026) ([Link to List of mentors](#))
 - Mentor, Quantum Computing Hackathon Pakistan (2026, 2nd Runner-Up) ([Webpage Link](#))
 - My work as research engineer at NSTIP lead to a new course development on Model based Systems Engineering, where I prepared tutorials and case studies.

Awards

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- First Prize in QInnovation World Challenge by Thales at Quantum Innovation Summit 2025, Dubai, UAE ([Challenge Link](#))
 - Best Researcher Award, eBRAIN Lab, NYUAD, UAE (2024–2025)
 - Fully-funded Ph.D. scholarship, Hamad Bin Khalifa University, Qatar (2020–2023)
 - Fully-funded M.Sc. scholarship (TA/RA), Istanbul Şehir University, Turkey (2018–2019)
 - Outstanding Performance Award, NSTIP, Saudi Arabia (2015–2017)

Professional Service and Leadership

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- **Reviewer**: IEEE Transactions on Emerging Topics in Computing, Advanced Quantum Technologies (Wiley), Engineering Applications of Artificial Intelligence (Elsevier), Computers & Electrical Engineering (Elsevier), Expert Systems with Applications (Elsevier), Journal of Computational Physics (Elsevier), Neurocomputing (Springer), SoftwareX (Elsevier), IEEE Quantum Week (QCE), IJCNN.
 - **TPC Member**: 2026 IEEE International Workshop on Quantum Computing: Circuits, Systems, Automation, and Applications (QC-CSAA) Co-located with ISVLSI 2026.
 - Co-organized special sessions and workshops at leading conferences, including IEEE IJCNN and IEEE Quantum Week (QCE).
 - Co-organized “*Quantum Federated Learning for Fraud Detection*” challenge in Quantum Innovation summit 2025, Dubai, UAE ([Challenge Link](#))

Invited Talks and Conference Presentations

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- ACM/IEEE Design Automation Conference (DAC), San Francisco, USA (June 2025)
 - IEEE Quantum Week (QCE), New Mexico, USA (Sep 2025)

- IEEE International Conference on Rebooting Computing (ICRC), San Diego, USA (Dec 2024)

Technical Skills

- **Quantum & QML:** PennyLane, Qiskit, IBM Quantum backends, hybrid QNNs, noise modeling, robustness, trainability/barren plateaus, expressibility analysis
- **ML & Software:** Python, NumPy/SciPy, scikit-learn, Keras/TensorFlow; benchmarking;
- **Optimization:** Resource-aware NAS, analytical/hardware-calibrated cost modeling, circuit depth/two-qubit gate minimization
- **Additional Skills:** C/C++, MATLAB; Verilog/VHDL; FPGA/Vivado

Selected Publications (*Complete publication list available on Google Scholar*)

Journals

- J6 **M. Kashif** and S. Al-Kuwari, “ResQNNs: A Residual Approach for Mitigating Barren Plateaus in Quantum Neural Networks”, *EPJ Quantum Technology*, 2024. <https://doi.org/10.1140/epjqt/s40507-023-00216-8>.
- J5 **M. Kashif** and M. Shafique, “Deep quantum neural networks with enhanced trainability and gradient propagation”, *Nature Scientific Reports*, 2025, <https://www.nature.com/articles/s41598-025-06035-4?>
- J4 T Ahmed, **M Kashif**, A Marchisio, M Shafique, Quantum Neural Networks: “A Comparative Analysis and Noise Robustness Evaluation”, *Nature Scientific Reports*, 2025. <https://www.nature.com/articles/s41598-025-17769-6>
- J3 **M. Kashif** and S. Al-Kuwari. “The impact of cost function globality and locality in hybrid quantum neural networks on NISQ devices”, *Machine Learning: Science and Technology*, 2023. <https://iopscience.iop.org/article/10.1088/2632-2153/acb12f/meta>.
- J2 **M. Kashif** and S. Al-Kuwari, “The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs”, *International Journal of Parallel, Emergent and Distributed Systems*. <https://www.tandfonline.com/doi/full/10.1080/17445760.2023.2231163>
- J1 **M. Kashif** and S. Al-Kuwari, “Physical Realization of Measurement Based Quantum Computation,” *IEEE Access*, 2023. <https://ieeexplore.ieee.org/abstract/document/10160024/>.

Conferences

- C8 **M. Kashif**, A. Marchisio, M Shafique, “Closing the loop: Resource-aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling”, *Accepted* in IEEE/ACM Design Automation Conference (*DAC*), Long Beach, CA, USA, July 2026. [arXiv:2603.00625](https://arxiv.org/abs/2603.00625)
- C7 **M. Kashif**, M Shafique, “Late Breaking Results: Hardware-Aware Compilation Reshapes Trainability in Variational Quantum Circuits”, *Accepted* in IEEE/ACM Design Automation Conference (*DAC*), Long Beach, CA, USA, July 2026. arxiv.org/abs/2604.16527
- C6 **M. Kashif**, *et.al.*, “FAQNAS: FLOPs-aware Hybrid Quantum Neural Architecture Search using Genetic Algorithm”, *Accepted* in IEEE Design Automation and Test in Europe (*DATE*), Verona Italy, April 2026. <https://arxiv.org/abs/2511.10062>
- C5 **M. Kashif**, A. Marchisio, M. Shafique, “Computational Advantage in Hybrid Quantum Neural Networks: Myth or Reality?,” IEEE/ACM Design Automation Conference, (*DAC*), San Francisco, USA, June 2025. <https://doi.org/10.1109/DAC63849.2025.11132906>
- C4 **M. Kashif** and M. Shafique, “Hqnet: Harnessing quantum noise for effective training of quantum neural networks in nisq era”, IEEE International Conference on Quantum Artificial Intelligence, (*QAI*), 2025. <https://doi.org/10.1109/QAI63978.2025.00066>
- C3 **M. Kashif**, *et.al.*, “Alleviating barren plateaus in parameterized quantum ML circuits: Investigating advanced parameter initialization strategies,” IEEE Design Automation and Test in Europe, (*DATE*), Valencia, Spain March 2024. <https://doi.org/10.23919/DATE58400.2024.10546644>
- C2 **M. Kashif** and S. Al-Kuwari, “Qiskit As a Simulation Platform for Measurement-based Quantum Computation,” 2022 IEEE 19th International Conference on Software Architecture Companion (*ICSA-C*), Honolulu, HI, USA, 2022, pp. 152-159, <https://doi.org/10.1109/ICSA-C54293.2022.00037>.
- C1 N. Innan, **M Kashif**, A Marchisio, YS Gan, F Barbaresco, M Shafique, “QUAV: Quantum-Assisted Path Planning and Optimization for UAV Navigation with Obstacle Avoidance”, IEEE International Conference on Quantum Artificial Intelligence (*QAI*), 2025. <https://doi.org/10.1109/QAI63978.2025.00040>.